A 90.7-nW Vibration-Based Condition Monitoring Chip Featuring a Digital Compute-in-Memory-Based DNN Accelerator Using an Ultra-Low-Power 13T-SRAM Cell

Haochen Zhang, *Student Member, IEEE*, Wei-Han Yu, *Member, IEEE*, Zhizhan Yang, *Student Member, IEEE*, Ka-Fai Un, *Member, IEEE*, Jun Yin, *Senior Member, IEEE*, Rui P. Martins, *Life Fellow, IEEE*, and Pui-In Mak, *Fellow, IEEE*

***Abstract*— This article presents an energy-harvester-powered ultra-low power (ULP) vibration-based condition monitoring (VbCM) chip with a digital compute-in-memory (CIM)-based deep neural network (DNN) accelerator. The VbCM chip achieves end-to-end signal processing, including a piezoelectric (PZ)-based accelerometer sensor and its readout circuit (RoC), a digital CIM-based DNN accelerator and a ULP radio system. To perform the targeted rotating bearing anomaly detection task with best-in-class accelerator power consumption and energy efficiency, the chip integrates several technologies from the algorithm level to the hardware level, including: 1) Compressing deep neural network (C-DNN) to reduce the network size through a feature compressor module (FCM); 2) Signal chain characteristic adaptation from direct training of the time-domain feature extractor (TD-FEx); 3) ULP 13T SRAM CIM bitcell to perform high energy efficiency in-cell multiplication; and 4) ripple counter (RCNT)-based accumulation scheme to improve the overall energy efficiency of the accelerator under a 0.35 V supply voltage. The VbCM chip achieves 90.7 nW total power with an inference FR of 20 frames/s, and the DNN accelerator's energy efficiency achieves 18.8 fJ/MAC. With all the weight parameters statically stored in the CIM macros, the accelerator can operate with an inference accuracy of 91.3% while consuming 24.7 nW.**

***Index Terms*— 13T-SRAM, accelerometer sensor, compute-in-memory (CIM), deep neural network (DNN), feature extractor, Internet-of-Things, quantization, ultra-low power (ULP), vibration-based condition monitoring (VbCM).**

Manuscript received 24 November 2023; revised 2 May 2024; accepted 10 June 2024. Date of publication 24 June 2024; date of current version 20 January 2025. This article was approved by Associate Editor Sanu K Mathew. This work was supported in part by the Science and Technology Development Fund, Macau SAR, under Grant 004/2023/SKL; in part by the University of Macau Development Fund under Grant IME/UMDF/006/2021; and in part by the New Cornerstone Science Foundation through the XPLORER PRIZE. *(Corresponding author: Pui-In Mak.)*

Haochen Zhang, Wei-Han Yu, Zhizhan Yang, Ka-Fai Un, Jun Yin, and Pui-In Mak are with the State-Key Laboratory of Analog and Mixed-Signal VLSI, the Institute of Microelectronics, the Faculty of Science and Technology, and the Department of Electrical and Computer Engineering, University of Macau, Macau, China (e-mail: pimak@umac.mo).

Rui P. Martins is with the State-Key Laboratory of Analog and Mixed-Signal VLSI, the Institute of Microelectronics, the Faculty of Science and Technology, and the Department of Electrical and Computer Engineering, University of Macau, Macau, China, and also with the Instituto Superior Técnico, Universidade de Lisboa, 1049-001 Lisbon, Portugal.

Color versions of one or more figures in this article are available at https://doi.org/10.1109/JSSC.2024.3413891.

Digital Object Identifier 10.1109/JSSC.2024.3413891

I. Introduction

MACHINE-LEARNING (ML)-based time-series analysis has revolutionized a wide variety of human-interactive pattern-recognition applications at the edge, such as voice activity detection [1], [2], keyword spotting (KWS) [3], [4], and arrhythmia detection [5], [6]. The success paves the way for applying edge computation to industrial applications, such as vibration-based condition monitoring (VbCM) [7], [8], [9]. VbCM can continuously evaluate the healthiness of industrial machinery, providing real-time status information that allows predictive maintenance. As the application of VbCM can greatly reduce manufacturing downtime and labor costs, the efficiency of industrial production can be improved. However, to get comprehensive status information on the asset under monitoring, massive deployment of the VbCM nodes is a must. As a result, the nodes are required to keep fully functional with a sub-100 nW power budget to prevent frequent battery replacement [5] (e.g., ten years lifetime with a thin button cell battery) or even realize full energy autonomous with limited power from harvesting source [10]. Current solutions [7], [8], [9] fail to meet the power requirement since the solutions analyze the acquired data with cloud-based ML algorithms. Regular wireless transmission is thus necessitated, and the wireless transmission dominates the total power consumption of a VbCM node. In [7], raw time-series data are upstreamed with $>$10 mW power consumption. Brown et al. [8] and Lukas et al. [9] transform raw data into frequency coefficients with the FFT block and compress the payload in wireless transmission. However, they still consume >2 μW even with specifically designed radio blocks for ULP scenarios.

Moving the DNN algorithms from the cloud to the edge is a natural step, as wireless transmission activity can be significantly reduced. Wireless transmission is triggered only when the DNN outputs an anomaly classification result that lengths several bits. Two design spaces should be co-explored to deploy DNNs for the VbCM application at the edge. At the hardware level, the DNN accelerator should keep high energy efficiency with tens of nanowatts. At the algorithm level, the DNN model should remain compact in size with dense parameters to prevent excessive ON-chip memory footprint and

0018-9200 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

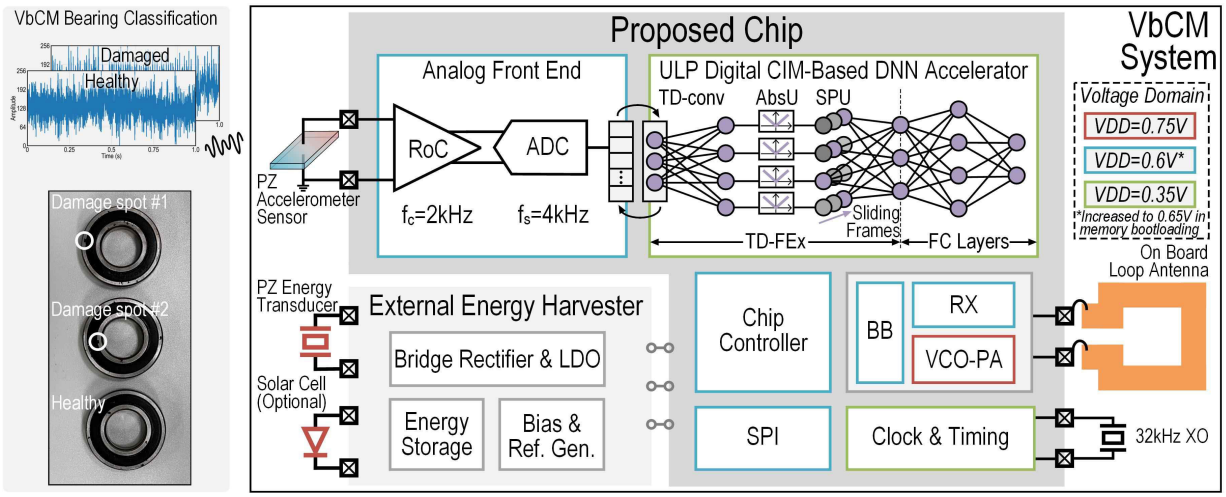


Fig. 1. Architecture of the proposed VbCM chip.

computation, which can also shrink the required computation resource and power on the chip. Although a number of ULP DNN accelerators have been proposed recently [1], [3], [4], [5], [6], they are not fit for the targeted application. For example, [5] and [6] both need μW-level power to process the temporal-dense machine vibration signal since they are targeted for temporal random sparse events (e.g., arrhythmia detection). Oh et al. [1] reduces its power consumption to $\sim$70 nW through extensive power gating. However, the size of the DNN is too small for the VbCM task. Under a given power budget, the size of the Von-Neumann computing architecture-based DNN accelerator is limited by the low computing energy efficiency, which consumes 1.5 pJ for each 1-bit multiply-and-accumulate (MAC) operation. Some works adopt compute-in-memory (CIM) techniques [11], [12], [13], [14] to improve the energy efficiency, yet they consume microwatt-level power consumption that exceeds nanowatt-level power budget. Tan et al. [4] explores the application of the CIM technique for a low-power KWS task, which achieves 5.1 fJ/MAC with 2.9 μW power consumption. Yet, it has a $>$320 nW leakage power that is still around 3$\times$ higher than the targeted power constraint for the VbCM application.

This work presents a VbCM chip that features a ULP digital CIM-based DNN accelerator to perform anomaly bearing detection. The chip also consists of an analog front-end (AFE) and a radio system for system completeness. With the vibration signal completely analyzed on the chip, wireless transmission activity can be significantly reduced, which enables a sub-100 nW long-term overall power consumption. A four-layer compressing deep neural network (C-DNN) directly performs inference on time-domain input signals. The time-domain feature extractor (TD-FEx) utilizes informative features by squeezing out zero-valued parts, compressing both the feature and network size. As the C-DNN can be trained to adapt to the in-band ripple of the sensor, we take advantage of a high-sensitivity ($\sim$4 mV/g) and low-price piezoelectric (PZ)-based accelerometer sensor to leverage both the AFE power and system cost. A ULP digital CIM-based accelerator is proposed to enable low-power and high-efficiency DNN inference. The accelerator stores all the weight parameters statically and computes all the layers in the C-DNN in parallel under a 0.35 V supply voltage. The CIM macros in the DNN accelerator are designed with a ULP 13T SRAM CIM bitcell that performs in-cell AND operation with low leakage power. The accelerator performs high-efficiency accumulation through a ripple counter (RCNT)-based accumulation scheme, which reduces the duration for each compute cycle and improves macro energy efficiency under ultra-low voltage scenarios.

The remaining part of this article is organized as follows. Section II describes the architecture of the proposed VbCM chip and the C-DNN. Section III details the circuit design of the proposed ULP 13T-SRAM bitcell and the RCNT-based accumulation scheme. Section IV reports the test settings and measurement results of the VbCM system. Section V concludes this article.

II. Proposed VbCM Chip

A. Architecture Overview

Fig. 1 shows the block diagram of the proposed VbCM system, where the AFE, the DNN-based data processing circuit, and the radio system are implemented on the VbCM chip. The signal chain of the chip starts with an accelerometer sensor, which is a lead zirconate titanate-based PZ sheet that works in compression mode along the z-axis. The vibration signal sensed by the accelerometer sensor is transduced as voltage output and is further amplified by the readout circuit (RoC) with a bandwidth of 2 kHz. The RoC consists of an LNA and a PGA that are cascaded to balance the noise and gain. An 8-bit SAR ADC sampling at 4 kHz directly digitizes the output of the RoC to a ping-pong buffer with a depth of 60 samples for the digital CIM-based DNN accelerator (detailed in Section III) to compute. The accelerator statically stores all the 1.4 KB weight parameters of C-DNN in 48 CIM macros, eliminating weight movement during its operation. The first layer of the C-DNN acts as a TD convolution-based FEx with a feature compressor module (FCM) (detailed in

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

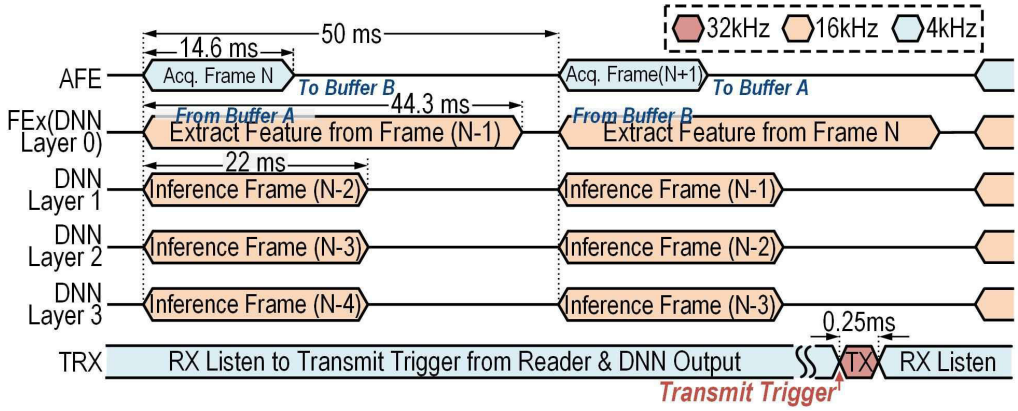


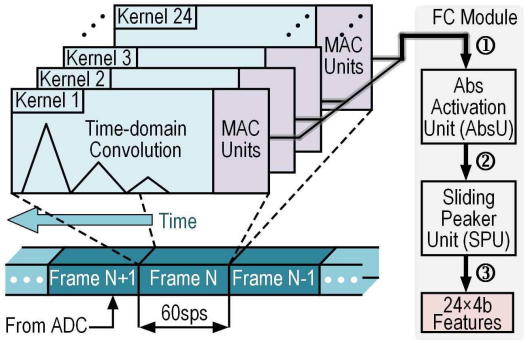
Fig. 2. Timing diagram of the proposed VbCM chip.

Section II-B). The 24 output features are processed by three fully connected (FC) layers with 30, 15, and 2 neurons separately. The C-DNN classifies the health condition of a rotating bearing from either healthy or anomaly conditions. The accelerator works at 16 kHz with a frame rate (FR) of 20 and a minimum inference latency of 250 ms. A ULP radio system working in 900 MHz ISM band [15] is implemented to realize wireless access to the VbCM node, which adopts a power-efficient VCO-PA architecture that reuses the inductor in the high-Q resonant tank as the antenna. The inference results are transmitted upon the system detecting an anomaly from the C-DNN. The results can also be accessed while the wake-up receiver (WuRX) in the radio system recognizes a wake-up signature. Fig. 2 shows the timing diagram of the aforementioned circuit blocks. The AFE (including the RoC and the ADC) and each layer in the DNN (including the FEx) are all working in parallel and form a five-stage pipeline, leading to a minimum VbCM latency of 250 ms. All the operations in the VbCM chip are orchestrated by a central controller with a 32 kHz crystal oscillator.

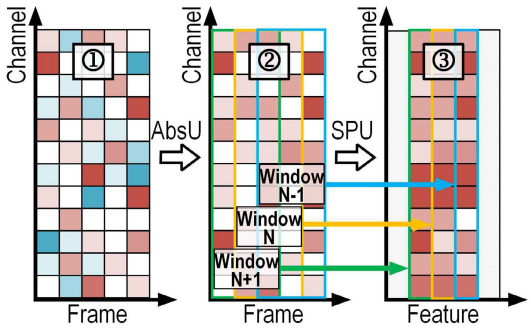
B. Compressing Deep Neural Network

As shown in Fig. 3(a), the proposed C-DNN directly takes 60 time-domain samples from the ping-pong buffer as the input. The samples will cover the frequency from 67 Hz to 2 kHz, which includes most of the vibration information from a rotating bearing needed for analysis [16]. Time-domain convolutions are performed between the input and 24 kernels, and the kernels are trained to adapt to both the sensor characteristics and the VbCM application. All the activations and the weights are with 4-bit quantization [17]. The proposed C-DNN has an inference accuracy of 90.2% and self-built dataset (detailed in Section IV-A). The C-DNN also exhibits a comparable accuracy of 91.4% when tested on a second open-source dataset [18]. As a comparison, [19] and [20] have a ~3% higher accuracy for 25× memory footprint and 90× floating-point computations, which are unlikely to be practical for extreme-edge deployment.

1) Feature Compressor Module: Fig. 3 shows the architecture and operations of the TD-FEx and the proposed FCM. The FCM cascades an absolute-value function activation unit



(a)



(b)

Fig. 3. (a) Architecture of the first layer of the C-DNN, which consists of a TD-conv layer and a FCM. (b) Operation diagram of the proposed FCM.

(AbsU) and a sliding peaker unit (SPU) to process MAC results by the CIM macros. Compared with the TD-FEx in [2] and [4] for processing time-varying voice signals, the proposed TD-FEx with AbsU and SPU is better at handling stationary machine vibration signals. Traditional rectified linear unit (ReLU)-based activation function used in [2] and [4] rectifies all the negative correlation results to zero, which only extracts the similarity information between the input and the kernels. The AbsU, on the other hand, additionally retrieves magnitude information hidden in the negative time-domain correlation result, which represents synchronism between the input and an out-of-phase kernel. The SPU following the AbsU continuously compares and outputs the feature with the largest

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

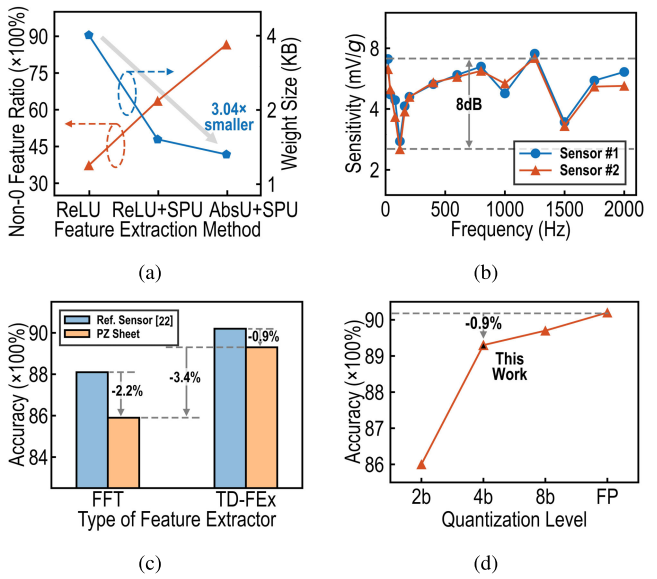


Fig. 4. (a) Nonzero feature ratio in an ablation study with different feature extraction methods and the weight size with corresponding methods under the same classification accuracy. (b) Measured sensitivity of the PZ-based accelerometer sensor with 8 dB in-band ripple. (c) Inference accuracy of FFT-based FEx and the proposed TD-FEx. (d) Inference accuracy comparison with different quantization levels in electrical test.

value in each channel with a sliding time window. The SPU expands the reception field of the FEx and fuses informative features in adjacent frames. The size of the sliding window is set to 3 to balance the latency of the C-DNN and the reception field of the FEx. An ablation study experiment is conducted to verify the effectiveness of AbsU and SPU. As shown in Fig. 4(a), the ratio of nonzero activation is increased by 26.3% though applying the SPU to the ReLU baseline, and the ratio can be further improved by 23% if we replace the ReLU function to AbsU. With the extra information extracted by the FCM, the proposed C-DNN can achieve the same accuracy with ∼3× more compressed network size than the ReLU baseline. As only several extra comparisons are needed in the SPU, the operations in the FCM do not require additional processing power for MAC, which results in roughly one-third of accelerator power consumption compared to the ReLU baseline.

2) TD-FEx Adaptation to Sensor Characteristics: In this design, we utilize a PZ sheet to transduce machine vibration into voltage output. As improving the sensor sensitivity enables reduction of RoC power though relaxed noise requirement [21], we increase the sensitivity of the accelerometer sensor to ∼4 mV/g [Fig. 4(b)] through augmenting its size to $10 \times 10 \times 1$ (unit: mm). The sensitivity corresponds to an input referred noise requirement of the RoC to 60 μV, which is achieved by the 24.7 nW RoC with an inverter-based LNA for first-stage signal amplification. However, the in-band frequency response of the sensor exhibits a ∼8 dB ripple, which results from various reasons, including mechanical impedance mismatching, sensor material defections, and complex vibration modes from increased dimension. These compound nonidealities can nonetheless be calibrated through training the DNN with the raw output of the AFE. As shown in Fig. 4(c), the TD-FEx outperforms the traditional FFT-based FEx in terms of inference accuracy by 3.4% because the TD-FEx can be an analogy to a set of trainable finite impulse response filters with more degree of sensor and application characteristics adaptation. Compared with the inference accuracy using the dataset that is acquired with a reference commercial sensor [22] with <0.1% in-band frequency ripple and >2000× price, the C-DNN equipped with TD-FEx only drops by 0.9%, whereas the FFT-based DNN drops by 2.2%. The 4-bit quantization [17] exhibits a 0.9% accuracy drop when compared with the floating-point model as shown in Fig. 4(d).

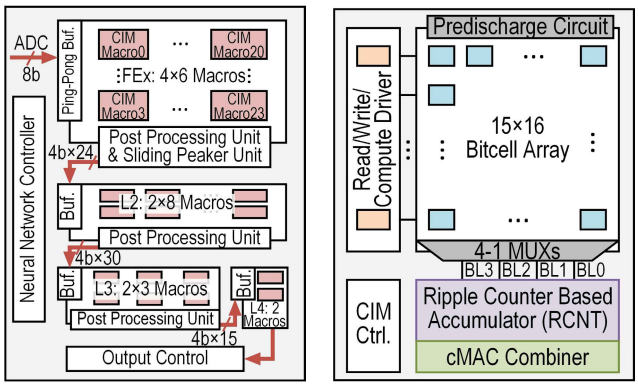


Fig. 5. (a) Architecture of the digital CIM-based DNN accelerator. (b) Schematic of the CIM macro.

III. ULP Digital CIM-Based DNN Accelerator

Fig. 4(a) shows the ULP digital CIM-based DNN accelerator that works under a 0.35 V supply voltage with a leakage power of 8.9 nW. The accelerator stationarily stores all the weights of the four-layer tiny DNN in 48 CIM macros, and the energy consumption for the initial bootloading varies with different storage mediums (∼0.23 μJ for DRAM [23] and ∼8.3 μJ for EEPROM [24]), which prevents energy consumption of frequent weight movement. As shown in Fig. 4(b), each macro consists of an array of 13T-SRAM bitcells with 15 wordlines (WLs) and 16 bitlines (BLs) and stores 60 unsigned 4-bit weights. All four layers in the C-DNN are mapped to the accelerator with 24, 16, 6, and 2 macros separately. Each layer contains an input buffer and a postprocessing unit (PPU), enabling fully parallel operation across each layer. The PPU performs partial-sum accumulation, signed-computation extension, normalization, and quantization. An output control block stores inference results with a programmable depth, and the final inference result is determined based on a majority voting scheme, which improves the inference accuracy at the cost of increased classification latency.

A. Ultra-Low-Power 13T-SRAM Bitcell Design

Fig. 6 shows the schematic of the proposed 13T-SRAM bitcell and the read/write/compute (R/W/C) WL driver. The bitcell consists of one cross-coupled inverter-based latch for

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

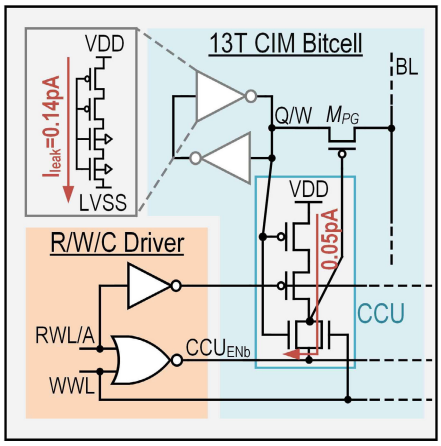


Fig. 6. Schematic of the proposed 13T-SRAM CIM bitcell and the read/write/compute driver.

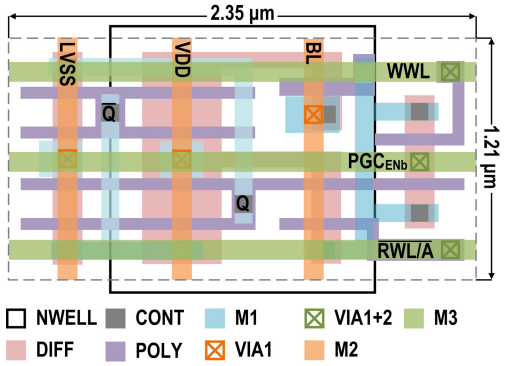
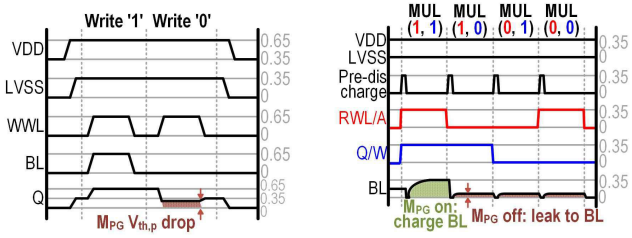


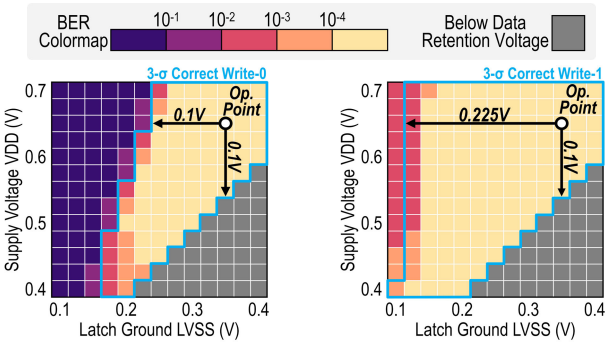
Fig. 7. Layout of the proposed 13T-SRAM CIM bitcell.



(a) (b)

Fig. 8. (a) Operation waveform of the proposed bitcell in write mode. (b) Operation waveform of the proposed bitcell in compute and read mode.

data storage, one PMOS access transistor M_{PG}, one BL, and one compute-and-control unit (CCU) to control the on/off state of M_{PG} as well as in-cell bitwise AND computation. The width of the pull-up transistors is increased to 400 nm to ensure the accelerator is working at 16 kHz and 20 frames/s. Unlike previous SRAM bitcells working in ultra-low voltage that utilizes at least two BLs [25], [26], [27], the proposed bitcell adopts single BL structure to perform both write and read operations, which leads to a compact bitcell area of 673 F^2 (Fig. 7). The single BL topology saves ~30% bitcell area and ~50% read energy (3.5 fJ/bit in simulation) compared with the bitcell that adds a complementary BL. A one-time write-assist procedure is required when bootloading the weights to the SRAM, which shifts up both the power rail (VDD) and the



(a) (b)

Fig. 9. 10 000-point Monte-Carlo simulations of (a) write "0" BER of the proposed bitcell and (b) write "0" BER of the proposed bitcell.

ground rail for the core latch (LVSS) to ensure successful write operations. To suppress leakage current, the cross-coupled latch in the bitcell is designed with two-layer-stacked inverters, with each having a 0.14 pA measured leakage current. The CCUs in the same row of the array are all power gated by the NOR gate in the R/W/C driver, which also suppresses the leakage current of the CCU to 0.05 pA.

To initialize the write operations, the power rail (VDD) of the SRAM is first increased to 0.65 V [Fig. 8(a)]. The ground rail of the latch (LVSS) is increased to 0.35 V afterward, leaving the ground rails for all other circuits at 0 V. It is worth noting that the PMOS M_{PG} is only able to pass a full logic "1" due to the existence of a $V_{th,p}$ drop, and this can be nonetheless solved by the up-shifted LVSS power rail, since the latch finds the logic "0" to the raised 0.35 V. In this write-assist voltage domain, the pull-down NMOS transistors are weakened by the reversed body biasing, and the whole latch is further weakened by the reduced 0.3 V VDD-LVSS voltage stress. Once the write-assist power rails settle down, M_{PG} can write the data signal on the BL to the latch. When the WWL signal in the targeted row-to-write is pulled high, the CCU_{ENb} signal is pulled down, and the CCU closes the M_{PG} for writing the data. Upon finishing the write operation, the write assist exits by sequentially dropping LVSS and VDD to 0 and 0.35 V. The write assist process necessitates that the VDD rail's up-shifting period covers the LVSS rail's up-shifting period, ensuring that the voltage stress on the core latches remains above the data retention voltage (DRV).

To verify the write performance of the proposed bitcell with up-shifted supply rails, we performed 10 000-point Monte-Carlo simulations with different VDD and LVSS combinations. Fig. 9(a) and (b), respectively, shows the bit error rate (BER) contour for writing the "0" operation and writing the "1" operation. The write-assist voltage rails ensure a >0.1 V voltage safety margin for 3-σ percentage of successful bit-level write operations. Two additional 1-million-point Monte-Carlo simulations are performed to show a $<10^{-6}$ write BER at the proposed write assist rails (0.65 V for VDD and 0.35 V for LVSS).

During the compute operations, all supply voltages and ground rails are separately kept to 0.35 and 0 V [Fig. 8(a)]. Each compute cycle starts with discharging all the BLs to 0 V.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

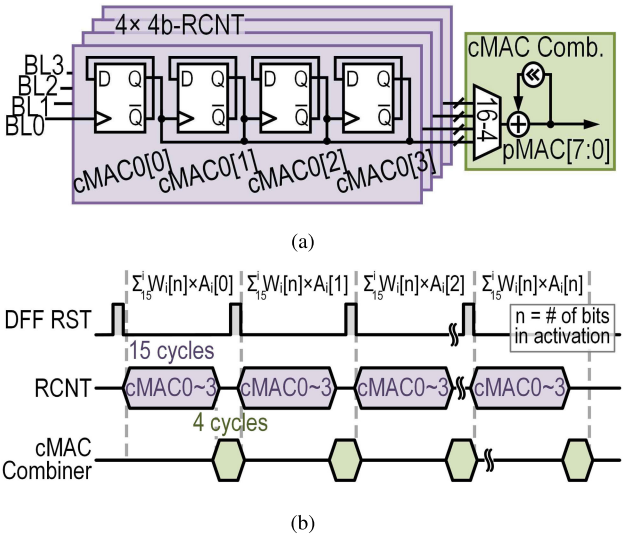


Fig. 10. (a) Schematic of the RCNT-based accumulator. (b) Timing diagram of the RCNT-based accumulation scheme to perform MAC operation between 15 1-bit activation and 4-bit weight.

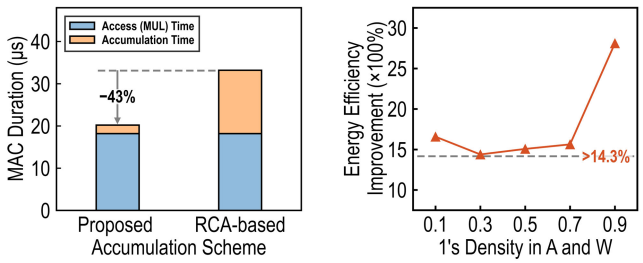


Fig. 11. (a) Comparison of MAC operation duration between the proposed RCNT-based accumulator and the traditional RCA. (b) Computation energy efficiency improvement of RCNT-based accumulator over RCA-based accumulator with different number distributions in activations and weights.

The RWL/A signal connecting to the input activation buffer in the targeted row to read is then pulled high. As a result, the CCU_{ENb} signal is pulled low through the NOR gate in the R/W/C driver, providing the CCU with a pull-down path. The CCU forms an inverter, which takes the data stored in the latch as the input and controls the M_{PG}. The M_{PG} is closed if the latch stores a "1," and the BL is charged high. In contrast, if a "0" is stored in the latch, M_{PG} keeps open in the compute cycle, and the voltage on the BL is kept to be the discharged "0" state. Therefore, RWL/A and data Q perform a bitwise AND multiplication on the BL, which is processed by the succeeding accumulators.

The read operation of the proposed bitcell can be seamlessly extended from its compute operation. In the read mode, the RWL/A signal is connected to the read decoder rather than the input activation buffer, and pulling high the RWL/A signal will present the data Q on the BL. Since the bitcell does not connect to the BL while reading a "0," it only sees a read "1" disturb in read operations. Monte-Carlo simulations show that the read "1" operation exhibits a less than 20 mV read disturb, which could not flip the stored data in the latch. The read BER is below $<10^{-4}$ when sweeping the VDD value from 300 to 400 mV in 10 000-point Monte-Carlo simulations.

B. Ripple Counter-Based Accumulation Scheme

To improve the energy efficiency of the accumulator operation under ultra-low voltage, we adopt digital RCNTs to accumulate multiplication results [Fig. 10(a)]. As a comparison, analog-computing-based CIM counterparts [4], [6] cannot work under such a low supply voltage. The accumulator takes four 4-1 MUXs to select four BLs corresponding to one 4-bit weight, and each BL is followed by a 4-bit RCNT that consists of a chain of 4 D-flip-flops (DFFs). The DFFs are all reset to zero at the beginning of the computation. As the CIM controller sequentially selects one of 15 WLs per cycle, one of 15 4-bit products between a 1-bit activation and a 4-bit weight is presented on four selected BLs. Each of the four RCNTs simultaneously stores the 4-bit counter-MAC (cMAC) results according to the bit-significance of the 4-bit weight and increments the accumulation results once the BL is pulled up. It takes a total of 15 cycles to compute cMAC0 to cMAC3, which are then summed by a 4-bit-in 8-bit-out cMAC combiner to form an 8-bit partial-MAC (pMAC) result with four cycles [Fig. 10(b)]. This operation will repeat from LSB to MSB of the activation in a bit-serial manner. The pMACs corresponding to each bit in the activations are summed in the PPU to finish MAC computation between multibit activations and weights. The RCNT-based accumulator only takes one *Clock-to-Q* delay of the DFF to settle, whereas the traditional ripple carry adder (RCA)-based accumulators [14] need their carry to propagate through all the stages of the adder. Fig. 11(a) shows the postlayout simulation results of the MAC duration of an RCNT-based accumulator and an RCA-based accumulator, both in a 4-bit configuration. To perform the same 4-bit accumulation, the proposed RCNT-based accumulator only takes 2 μs to settle, whereas an RCA-based accumulator takes 15 μs. With a common duration for multiplication of 18.2 μs, the macro with the proposed RCNT-based accumulator has 43% less MAC duration than the RCA-based counterpart with <9% macro area overhead. Moreover, the MAC duration of the RCA-based accumulator exhibits a near linear increase with the number of accumulation operations, yet the RCNT-based structure keeps an unchanged one *Clock-to-Q* delay. As speeding up MAC duration reduces the contribution of leakage power to each computation, the proposed RCNT-based accumulator shows a >13% energy efficiency improvement [Fig. 4(b)] across different input activation and weight distributions.

IV. Measurements

The proposed VbCM chip is fabricated with TSMC 65-nm LP CMOS technology. Fig. 12 shows the micrograph of the chip. The chip occupies an area of 2 mm^2, where 15.1% is the active region (Fig. 13). The VbCM chip is wire-bonded on a printed circuit board (PCB) with an ON-chip antenna, decoupling capacitors, and a flexible printed circuit cable for powering the chip and transmitting monitoring signals [Fig. 14(a) and (b)]. To complete the VbCM system, a second PCB integrates OFF-chip implemented modules,

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

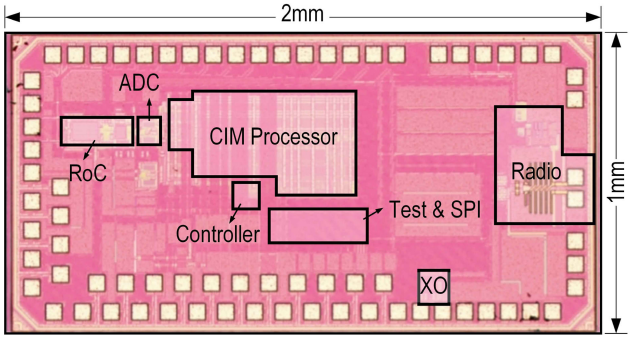


Fig. 12. Micrograph of the VbCM chip.

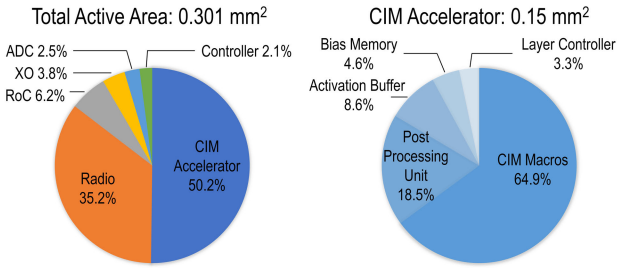
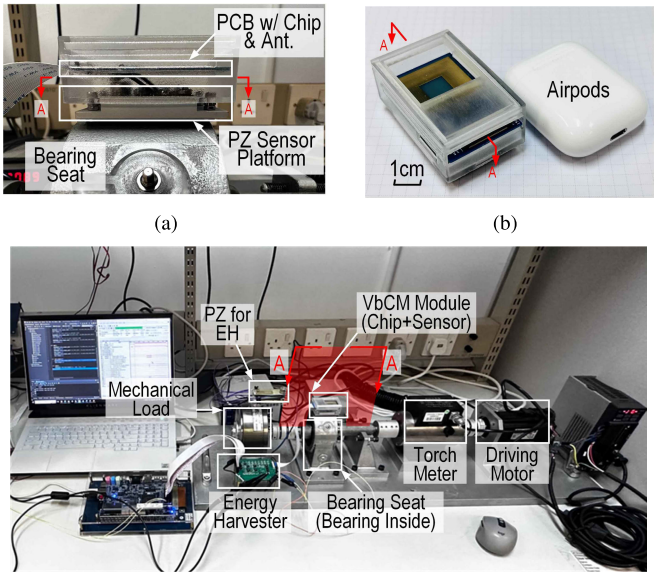


Fig. 13. Chip and CIM accelerator area breakdown.



(a) (b) (c)

Fig. 14. (a) Assembly of the VbCM system on the bearing seat on the rotating machine. (b) Photo of the VbCM module. (c) Test platform of the VbCM system.

including a PZ-based energy harvester, an optional solar energy harvester, low dropout voltage regulators, and bias generators. Level shifters that bridge the monitoring signals between the VbCM chip and the field-programmable gate array (FPGA) are also included on the second PCB, yet they are supplied by the FPGA. The whole test platform is shown in Fig. 14(c).

TABLE I

DATASET INFORMATION FOR ROTATING BEARING-BASED MACHINE HEALTH ANOMALY DETECTION

Dataset Information				
Class	Damage Type	Bearing Code	Sample Rate	Length/Bearing
1 (Healthy)	No Damage	ND01~ND05	4kHz	2* × 3** × 16s
2 (Damaged)	Inner Ring	IR01~IR08		
	Outer Ring	OR01~OR08		

* each data item is recorded with both reference sensor [22] and proposed sensor
** each data item is recorded under motor speed (unit: rpm) of 600, 900 and 1200

A. Test Platform and Dataset

To fully explore the effectiveness of the proposed system, we built a customized rotating machine-based bearing health monitoring platform as shown in Fig. 14(c). A driving motor is connected to a mechanical load through a shaft. The VbCM chip and the sensor are placed within a 3-D printed box, which is screwed on the bearing seat that holds a replaceable bearing. A PC-controlled software-defined radio (SDR) system is used to emulate an RFID reader that receives and decodes the signal sent from the VbCM chip at an indoor distance of 10 m. The host PC sends the weight data and chip configuration signals to the VbCM chip through the Python interface.

As shown in Table I, a customized dataset is built based on the test platform. The dataset records vibration signals from healthy or anomaly bearings as two classes to classify from. To enhance the applicability of the dataset, the anomaly class includes both inner-ring-damaged bearings and outer-ring-damaged bearings. Moreover, the rotation speed of the motor is set at three speeds, which are 600, 900, and 1200 r/min, separately. There are, in total, five bearings with no damage (numbered ND01-ND05), eight bearings with inner-ring damage (numbered IR01-IR08), and eight bearings with outer-ring damage (numbered OR01-OR08). Each bearing has its vibration signal recorded in a 16-s clip at 4 kHz by both the PZ-based sensor and a reference sensor. For the PZ-based sensor, the data are directly collected at the output of the ADC and trained. Direct training with ADC output enhances the accuracy of the system, as all the nonidealities along the signal path are included in training data. For the reference sensor, the data are collected by its corresponding evaluation board, followed by filtering and down-sampling with the Python Numpy library, which matches the sampling frequency and bandwidth to the signal acquired with the PZ-based sensor. To summarize, each sensor has 67 200 bearing vibration clips, and each clip has a length of 60 sample points at 4 kHz.

B. Performance Characterization

Fig. 15 shows the power breakdown of the proposed VbCM system operating at 20 frames/s with no TX activation, defining the power consumption's lower bound. The RoC has a total gain from 28 to 43 dB, where 20 dB is contributed by the first-stage inverter-based capacitor-feedback LNA. The total measured input-referred noise voltage is 35.1 μV, which is lower than the 60 μV noise requirement. The RoC consumes 24.6 nW to amplify the vibration signal. An 8-bit ADC further

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

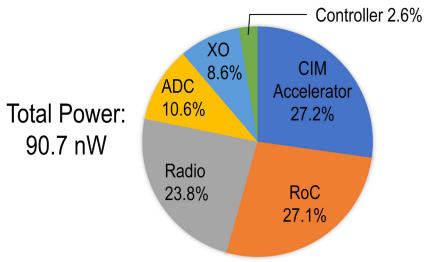


Fig. 15. Measured power breakdown of the VbCM chip.

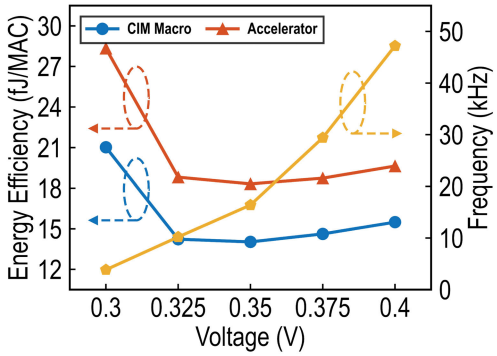


Fig. 16. Measured energy efficiency of the proposed CIM macro and overall accelerator.

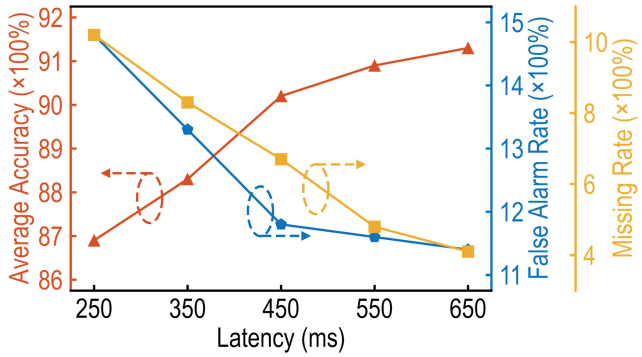


Fig. 17. Inference accuracy, FAR, and MR with respect to inference latency in the mechanical test.

consumes 9.6 nW to digitize the output of the RoC, which sums the total power consumption of the AFE to 34.2 nW under a 0.6 V supply voltage.

The digital CIM-based accelerator consumes 24.7 nW to perform the classification task. Operating under a supply voltage of 0.35 V, the energy efficiency of the whole DNN accelerator is 18.8 fJ/MAC, while an energy efficiency of 14.2 fJ/MAC is achieved at the CIM macro-level. Fig. 16 shows the voltage scaling of the energy efficiency and operating frequency of both the accelerator and the macro, demonstrating the proposed accelerator's potential to adapt to edge tasks with different power and throughput requirements. The functionality test for the proposed CIM processor is performed on five different chips, which all show correct computation output with the VDD ranging from 0.3 to 0.4 V. Fig. 17 shows the mechanical inference results. With the size of the majority voting window of 9, an overall accuracy of 91.3% is achieved. The corresponding false alarm rate (FAR) and missing rate (MR) are, respectively, 11.4% and 4.1%.

The RX of the ULP radio system is always on for the transmission request from the SDR. It consumes 21.6 nW while listening to the 4 kbps request signature. The RX has a sensitivity of −60 dBm in an over-the-air test. On the TX side, a 21.9% dc-to-RF efficiency is achieved by applying the VCO-PA architecture, which leads to a TX transmission efficiency of 0.23 nJ/bit.

C. ULP NN Accelerator Performance Benchmarking

Table II benchmarks our digital CIM-based DNN accelerator with the state-of-the-art ULP DNN accelerators. This work has first demonstrated the application of digital CIM techniques in nanowatt-class DNN accelerators under an ultra-low 0.35 V supply voltage. The accelerator outperforms the previous ULP DNN accelerators in terms of total power consumption, energy efficiency, and leakage power. Besides, the CIM accelerator benefits from the compact size of the proposed 13T-SRAM bitcell, leading to a total accelerator area of 0.151 mm^2. Compared to [5], the accelerator shows 52× better energy efficiency and 15× less power consumption. The proposed accelerator also exhibits 87× better energy efficiency and 3.2× less power consumption than [1]. Shan et al. [3] carefully optimized the memory and computation circuit with a 0.41 V supply voltage, yet its energy efficiency is still >40× less than our accelerator. For CIM-based ULP DNN accelerators, [4] suffers from a 36× larger leakage power of 320 nW and a 0.9 V supply voltage. Despite achieving slightly higher energy efficiency, the accelerator is unsuitable for the targeted VbCM application due to its large power consumption.

D. System Power Analysis

The total power consumption for the VbCM chip can be divided into two parts when processing the vibration signal at a fixed FR: a constant part for signal acquisition, processing and system controlling, and a dynamic part for TX transmission (P_{TX}). P_{TX} varies depending on: 1) the transmission mode of the TX and 2) the healthy status of the bearing-under-monitoring. Considering η_{TX} as the energy consumption for each transmission, P_{TX} is determined by the FAR while the bearing is under healthy condition

$$P_{TX} = \eta_{TX} \times \text{FR} \times \text{FAR} \tag{1}$$

whereas if the bearing is damaged, P_{TX} is determined by the MR

$$P_{TX} = \eta_{TX} \times \text{FR} \times (1 - \text{MR}). \tag{2}$$

Fig. 18 shows the system power consumption level with different TX activation intervals. The lower bound power consumption for the VbCM chip is 90.7 nW, which corresponds to the case where TX transmission is activated by an occasional external wake-up signature. The power will increase to 127.5 nW if all the inference results are transmitted through the TX. The VbCM chip consumes 94.9 nW with a healthy bearing and 125.6 nW with a damaged bearing.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PERFORMANCE SUMMARY AND COMPARISON OF THE PROPOSED ULP NN ACCELERATOR

ULP NN Accelerator	This Work	[5] ISSCC'21	[3] JSSC'20	[4] JSSC'23	[1] ISSCC'19	[6] ISSCC'22
Technology (nm)	**65 LP**	180	28	28	180	180
Operating Voltage (V)	**0.35**[a]	0.6	0.41	0.9	0.6	1.2 & 0.6
Architecture	**Digital CIM**	Von-Neumann	Von-Neumann	Analog CIM	Von-Neumann	Mixed-Signal CIM
Application	**VbCM**	KWS / ECG	KWS	KWS	VAD	ECG
Accuracy	**91.3%**	95%~99% @ ECG 85%~94% @ KWS	97.3% @ 1 word 94.6% @ 2 words	90.9%	91.5% @ Speech 90% @ Non-speech	90.5% @ ECG
Memory Cell Type	**13T ULP SRAM**	Customized Latch	Tri-state Latch	5T SRAM	Customized SRAM	Customized Latch
Bitcell Area (F^2)	**673**	14909[b]	1122	603	-	-
Bitcell Active Leakage Power (fW/bit)	**115**	5000	2700	4745	-	-
Memory Capacity (kb)	**11.52**	12.80	5.2	67.44	128	8.96
Frequency (kHz)	**16.384**	17	40	100	700 (Duty-cycled)	Clock-Free
Computation/Frame (kMACs)	**65.52**	1900	3.61	288	24	-
# of Network Parameters[b] (×1k)	**2.64**	1.51	3.456	12.40	1.568	2.24
Weight Bitwidth[b]	**4**	8	1/8	4	4	4
FPS	**20**	0.2	62.5	2	2	No Frame
Active Power (nW)	**24.7**	<378[d]	173	2910	78.9[c]	<350
Leakage Power (nW)	**8.9**	-	138	320	-	82
Energy Efficiency (fJ/MAC)	**18.8**	<994[c]	765[c]	10.2	1644[c]	530 fJ/SOP
Throughput (kMACs/s)	**1310**	380	226	28800	48	-
Active Area (mm^2)	**0.151**	7.5[e]	0.035[e]	0.048[f]	3.8[e]	3.5[e]

[a] 0.65V one-time write-assist for memory boot loading
[b] Network Size = # Network of Parameters × Weight Bitwidth
[c] calculated from data in the paper
[d] including a level-crossing ADC
[e] estimated from chip micrograph
[f] macro area excluding post-processing circuits, controllers and input buffers

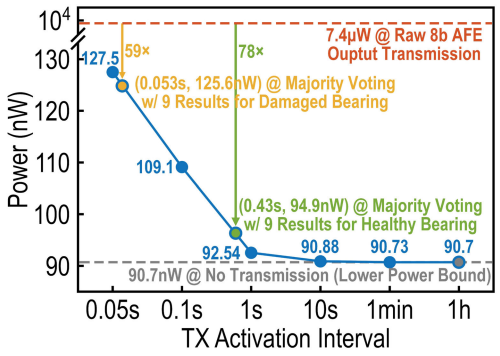


Fig. 18. System power consumption with different TX activation intervals.

Compared to IoT nodes that cannot process data locally, our proposed chip can greatly reduce node power consumption due to reduced wireless data transmission, which consumes >20× less power than [9]. The energy consumed on the VbCM chip can be completely covered by a vibration energy harvester (MIDE PPA-4011) with a simple full bridge rectifier that harvests 221 nW on our machine test platform with an outer-ring damaged bearing rotating at 600 r/min. Raw-data-based transmission will consume >59× higher power than inference result-based transmission, and the corresponding 7.4 μW power consumption hinders such VbCM nodes from energy self-sustainable operations.

V. CONCLUSION

This article reports a tiny digital CIM-based DNN accelerator on the VbCM task. The design explores the application of ULP ML systems on extreme edge applications from algorithm and circuit co-design. The proposed C-DNN running on the accelerator compresses its size by improving the information in the features. The TD-FEx in the C-DNN exhibits adaptation to sensor characteristics, which leverages sensor performance for other system-level metrics such as RoC power and VbCM node cost. The proposed 13-T ULP SRAM CIM bitcell and the RCNT-based accumulation scheme improve the energy efficiency and lower the power consumption of the 0.35 V nanowatt-class CIM-based DNN accelerator. As the proposed ULP ML techniques drastically reduce wireless transmission power, the power requirement for VbCM nodes is greatly relieved, which enables massive deployment of such systems for ever-increasing industrial productivity improvement.

REFERENCES

[1] S. Oh et al., "An acoustic signal processing chip with 142-nW voice activity detection using mixer-based sequential frequency scanning and neural network classification," *IEEE J. Solid-State Circuits*, vol. 54, no. 11, pp. 3005–3016, Nov. 2019.

[2] F. Chen, K.-F. Un, W.-H. Yu, P.-I. Mak, and R. P. Martins, "A 108nW 0.8 mm^2 analog voice activity detector (VAD) featuring a time-domain CNN as a programmable feature extractor and a sparsity-aware computational scheme in 28nm CMOS," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 1–3.

[3] W. Shan et al., "A 510-nW wake-up keyword-spotting chip using serial-FFT-based MFCC and binarized depthwise separable CNN in 28-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 56, no. 1, pp. 151–164, Jan. 2021.

[4] F. Tan, W.-H. Yu, K.-F. Un, R. P. Martins, and P.-I. Mak, "A 0.05-mm^2 2.91-nJ/decision keyword-spotting (KWS) chip featuring an always-retention 5T-SRAM in 28-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 59, pp. 626–635, Feb. 2024.

[5] Z. Wang et al., "A 148-nW reconfigurable event-driven intelligent wake-up system for AIoT nodes using an asynchronous pulse-based feature extractor and a convolutional neural network," *IEEE J. Solid-State Circuits*, vol. 56, no. 11, pp. 3274–3288, Nov. 2021.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.

[6] Y. Liu et al., "An 82nW 0.53pJ/SOP clock-free spiking neural network with 40μs latency for IoT wake-up functions using ultimate-event-driven bionic architecture and computing-in-memory technique," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 372–374.

[7] *Voyager 3, Wireless Condition Based Monitoring Platform*. Accessed: Aug. 3, 2020. [Online]. Available: https://wiki.analog.com/resources/eval/user-guides/industrial_wireless_cbm_evaluation_tool_3axis

[8] J. K. Brown et al., "27.1 A 65nm energy-harvesting ULP SoC with 256kB cortex-M0 enabling an 89.1μW continuous machine health monitoring wireless self-powered system," in *Proc. IEEE Int. Solid-State Circuits Conf.*, Feb. 2020, pp. 420–422.

[9] C. J. Lukas et al., "15.2 A 2.19μW self-powered SoC with integrated multimodal energy harvesting, dual-channel up to -92dBm WRX and energy-aware subsystem," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 238–240.

[10] M. Alioto, *Enabling the Internet of Things: From Integrated Circuits to Integrated Systems*. Cham, Switzerland: Springer, 2017.

[11] X. Si et al., "15.5 A 28nm 64Kb 6T SRAM computing-in-memory macro with 8b MAC operation for AI edge chips," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 246–248.

[12] J.-W. Su et al., "16.3 A 28nm 384kb 6T-SRAM computation-in-memory macro with 8B precision for AI edge chips," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 250–252.

[13] Y.-D. Chih et al., "16.4 An 89TOPS/W and 16.3TOPS/mm^2 all-digital SRAM-based full-precision compute-in memory macro in 22nm for machine-learning edge applications," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, vol. 64, Feb. 2021, pp. 252–254.

[14] B. Yan et al., "A 1.041-Mb/mm^2 27.38-TOPS/W signed-INT8 dynamic-logic-based ADC-less SRAM compute-in-memory macro in 28nm with reconfigurable bitwise operation for AI and embedded applications," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 188–190.

[15] Z. Yang, J. Yin, W.-H. Yu, H. Zhang, P.-I. Mak, and R. P. Martins, "A ULP long-range active-RF tag with automatic antenna-interface calibration achieving 20.5% TX efficiency at -22dBm EIRP, and -60.4dBm sensitivity at 17.8nW RX power," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 30–32.

[16] *An Overview of Bearing Vibration Analysis*. Accessed: Oct. 15, 2022. [Online]. Available: https://www.schaeffler.com/remotemedien/media/_shared_media/08_media_library/01_publications/schaeffler_2/technicalpaper_1/download_1/vibration_analysis_en_en.pdf

[17] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Nov. 2018, pp. 2704–2713.

[18] C. Lessmeier, J. K. Kimotho, D. Zimmer, and W. Sextro, "Condition monitoring of bearing damage in electromechanical drive systems by using motor current signals of electric motors: A benchmark data set for data-driven classification," in *Proc. PHM Soc. Eur. Conf.*, 2016, vol. 3, no. 1, pp. 1–17.

[19] V. Pandhare, J. Singh, and J. Lee, "Convolutional neural network based rolling-element bearing fault diagnosis for naturally occurring and progressing defects using time-frequency domain features," in *Proc. Prognostics Syst. Health Manage. Conf. (PHM-Paris)*, May 2019, pp. 320–326.

[20] D. He, W. Guo, and M. He, "Bearing diagnosis accuracy comparison using convolutional neural network with time/frequency domain signals," in *Proc. Prognostics Syst. Health Manage. Conf.*, Oct. 2019, pp. 1–6.

[21] Y. Peng et al., "A 184nW, 121μg/vHz noise floor triaxial MEMS accelerometer with integrated CMOS readout circuit and variation-compensated high voltage MEMS biasing," in *Proc. IEEE Symp. VLSI Technol. Circuits*, Jun. 2022, pp. 84–85.

[22] *Datasheet: ADXL356 3-Axis MEMS Accelerometer*. Accessed: Aug. 11, 2023. [Online]. Available: https://www.analog.com/media/en/technical-documentation/data-sheets/adxl356-357.pdf

[23] M. Horowitz, "1.1 Computing's energy problem (and what we can do about it)," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2014, pp. 10–14.

[24] *Datasheet: M95160-R 16-Kbit Serial SPI Bus EEPROM With High-Speed Clock*. Accessed: Apr. 19, 2024. [Online]. Available: https://www.st.com/resource/en/datasheet/m95160-r.pdf

[25] B. H. Calhoun and A. P. Chandrakasan, "A 256-kb 65-nm sub-threshold SRAM design for ultra-low-voltage operation," *IEEE J. Solid-State Circuits*, vol. 42, no. 3, pp. 680–688, Mar. 2007.

[26] S. Hanson et al., "A low-voltage processor for sensing applications with picowatt standby mode," *IEEE J. Solid-State Circuits*, vol. 44, no. 4, pp. 1145–1155, Apr. 2009.

[27] M.-F. Chang et al., "A sub-0.3 V area-efficient L-shaped 7T SRAM with read bitline swing expansion schemes based on boosted read-bitline, asymmetric-VA sub-0.3 V area-efficient L-shaped 7T SRAM with read bitline swing expansion schemes based on boosted read-bitline, asymmetric-V_{TH} read-port, and offset cell VDD biasing techniques," *IEEE J. Solid-State Circuits*, vol. 48, no. 10, pp. 2558–2569, Oct. 2013.



Haochen Zhang (Student Member, IEEE) received the B.S. degree in information science and engineering from Southeast University, Nanjing, China, in 2019, and the M.S. degree in electrical and computer engineering from the University of Michigan, Ann Arbor, MI, USA, in 2021. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau, Macau, China.

His research interests include algorithm, circuit, and system design for extreme-edge AI applications.



Wei-Han Yu (Member, IEEE) received the Ph.D. degree from the University of Macau (UM), Macau, China, in 2018.

From 2019 to 2021, he was a Visiting Scholar at Muramnn Mixed-Signal Group, Stanford University, Stanford, CA, USA. He has been an Assistant Professor with the State-Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), UM, since 2021. His research interests include edge AI, in-memory computing, switched capacitor circuits, energy-harvested RF transceivers, and neural interfaces.

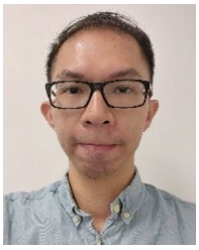
Dr. Yu received IEEE ISSCC Student Travel Grant Award, FDCT Science and Technology Postgraduate Student Award in 2016, and IEEE SSCS Predoctoral Achievement Award in 2018.



Zhizhan Yang (Student Member, IEEE) received the B.S. degree in electronic science and technology from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2016, and the M.E. degree in integrated circuit engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2019. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau, Macau, China.

His research interests include energy-efficient transceivers for wireless communications and ultra-low-power IC design.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.



Ka-Fai Un (Member, IEEE) received the Ph.D. degree from the University of Macau (UM), Macau, China, in 2014.

He was a Post-Doctoral Researcher and a Lecturer (Macao Fellow) in 2014 and 2015, respectively. He has been an Assistant Professor with the State Key Laboratory of Analog and Mixed-Signal VLSI, UM, since 2018. He was on leave from UM and was a Visiting Post-Doctoral Researcher with the School of Electrical and Electronic Engineering, University College Dublin, Dublin, Ireland, from 2017 to 2018. His research interests include analog artificial intelligence (AI) circuits, digital AI accelerator design, and analog and RF circuit design.

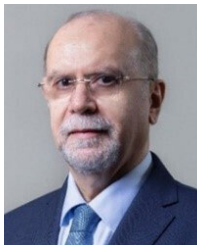
Dr. Un received Macau Mathematics Olympics and represented Macau from Chinese Mathematics Olympiad (CMO) and the International Mathematics Olympiad (IMO), in Changsha and Tokyo, in 2003, respectively. He was also a recipient of the 2008 APCCAS Merit Student Paper Certificate.



Jun Yin (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in microelectronics from Peking University, Beijing, China, in 2004 and 2007, respectively, and the Ph.D. degree in electronic and computer engineering (ECE) from Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2013.

He is currently an Associate Professor at the State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau (UM), Macau, China. His research interests include CMOS radio frequency (RF) integrated circuits and systems.

Dr. Yin is a TPC Member of IEEE International Solid-State Circuits Conference (ISSCC), IEEE Asian Solid-State Circuits Conference (A-SSCC), and IEEE European Conference on Solid-State Circuits (ESSCIRC). He serving as the Associate Editor for IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS— I: REGULAR PAPERS.



Rui P. Martins (Life Fellow, IEEE) received the Ph.D. degree in electrical engineering and computers from the Department of Electrical and Computer Engineering, Instituto Superior Técnico, Universidade de Lisbon, Lisbon, Portugal, in 1992.

He has been with the Department of Electrical and Computer Engineering, Instituto Superior Técnico, Universidade de Lisbon, since October 1980. Since 1992, he has been on leave with the Universidade de Lisbon and with the Department of Electrical and Computer Engineering, Faculty of Science and Technology, University of Macau, Macau, China, where he has been the Chair Professor since August 2013. He was the Founding Director of the State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau, from 2011 to 2022, where he is currently the Director of the Institute of Microelectronics. Since July 2010, has been an Academician with Lisbon Academy of Sciences, Lisbon. He has authored or coauthored more than 900 publications, including ten books, 12 book chapters, 50 patents, more than 300 articles in scientific journals, and more than 400 in conference proceedings. His research interests include analog and mixed-signal VLSI design.

Prof. Martins received the Author Recognition Award at the 70 years of ISSCC in 2023, a Top Contributor with more than 50 papers, and three Medals from Macau Government in 1999, 2001, and 2021.



Pui-In Mak (Fellow, IEEE) received the B.Sc. and Ph.D. degrees from the University of Macau (UM), Macau, China, in 2003 and 2006, respectively.

He is currently a Professor with the Faculty of Science and Technology–ECE, UM, the Director of the State-Key Laboratory of Analog and Mixed-Signal VLSI, UM, and the Deputy Director (Research) of the Institute of Microelectronics, UM. His research interests include analog and radio frequency (RF) circuits and systems for wireless and multidisciplinary innovations.

Dr. Mak is an Overseas Expert of the Chinese Academy of Sciences and a fellow of IET and U.K. Royal Society of Chemistry (RSC). He was a TPC Member of ISSCC, A-SSCC, and ESSCIRC. He received the Tencent Xplorer Prize in 2022 and was recognized in 2023 as one of the Top Contributing Authors of ISSCC in its 70 years of history. He is named as a Foreign Academician of Portuguese Academy of Sciences in 2024. He is currently the Editor-in-Chief of IEEE SOLID-STATE CIRCUITS LETTERS. He was a Distinguished Lecturer of the IEEE Circuits and Systems Society and IEEE Solid-State Circuits Society. His team reported 39 state-of-the-art chips at the 2011–2024 IEEE International Solid-State Circuits Conference (ISSCC).

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:23:04 UTC from IEEE Xplore. Restrictions apply.